



An official website of the United States government

[Here's how you know](#)

FULL TEXT LINKS



Begell House, Inc. Publishers

[Int J Med Mushrooms](#). 2014;16(3):247-57. doi: 10.1615/intjmedmushr.v16.i3.50.

Isolation and purification of a polysaccharide from the caterpillar medicinal mushroom *Cordyceps militaris* (Ascomycetes) fruit bodies and its immunomodulation of RAW 264.7 macrophages

Lina Zhu ¹, Qingjiu Tang ², Shuai Zhou ³, Yanfang Liu ², Zhong Zhang ⁴, Xinhua Gao ⁵, Shiping Wang ⁶, Zhaolong Wang ⁶

Affiliations

PMID: 24941166 DOI: [10.1615/intjmedmushr.v16.i3.50](#)

Abstract

A novel polysaccharide (CP2-S) was purified from *Cordyceps militaris* fruit bodies by hot water extraction, ethanol precipitation, DEAE-Sephacryl S-400 high-resolution chromatography. The polysaccharide had a molecular weight of 5.938×10^6 g/mol and was mainly composed of glucose. CP2-S had carbohydrate content estimated to be 100% using the phenol-sulfuric acid method. Immunostimulating experiments in vitro indicated that CP2-S could stimulate nitric oxide production, phagocytosis, respiratory burst activity, and secretion of interleukin-1 β and interleukin-2 of macrophages, suggesting that this water-soluble polysaccharide from the fruit body of *C. militaris* is a natural immunostimulating polysaccharide with potential for further application.

[PubMed Disclaimer](#)

LinkOut - more resources

Full Text Sources

[BegellHouse Publisher, Inc.](#)

Other Literature Sources

[The Lens - Patent Citations Database](#)

[scite Smart Citations](#)